

2026-06-21

Climatology prediction

<https://github.com/tglanz/climatology/tree/main/ml>

Context

Given a sequence of K **step diagnostic** ζ , instead of attempting to predict the state of the system at some later time t we attempt to predict a **climatology diagnostic** $[\bar{u}]$.

By doing so, we eliminate the stochastic dimension of the forcing (Stirring in our case).

Window Selection

- Window size K
- Start / End / Stride?

Climatology Range

What is the time frame to compute $[\bar{u}]$ for?

Learnability

The underlying heuristic is that the diagnostic evolution within a window determines the climatology

i.e. for different windows $W_{i,j}$

$$\textit{pred}(W_i) = \textit{pred}(W_j) = [\tilde{u}]$$

- We don't encode time in the model

Spinup and convergence

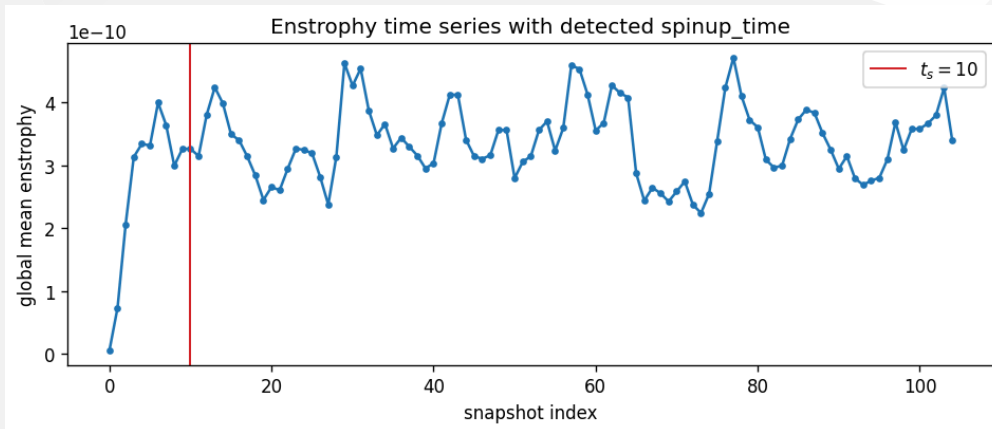
A simulation S is composed of M snapshots; Each snapshot holding dimensions (lons, lats, time, ...) and variables (vor, stirring, ucomp, ...).

- Windows are taken between the **Spinup Time** t_s and the **Convergence Time** t_c
- Climatology $[\bar{u}]_{t_c}^{t_M}$ is computed since t_c to the end of simulation

Spinup

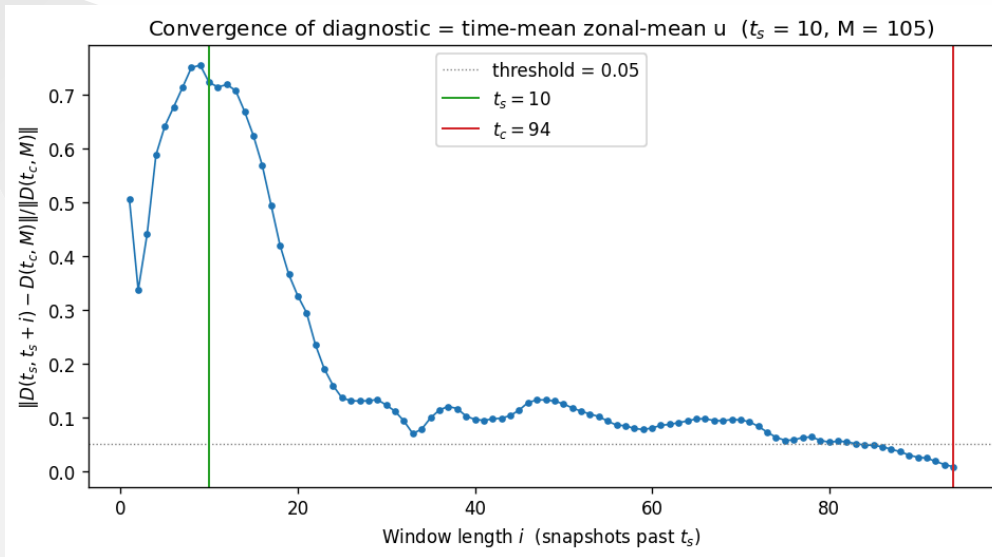
Spinup time is the time t_s after which the simulated dynamics have stabilized post initial conditions

t_c is determined by taking multiple subsequent enstrophy windows and thresholding their change of z-score



Convergence (1/2)

Convergence time is the time $t_c \geq t_s$ after which the time mean has settled similar to the global time mean



Convergence (2/2)

We determine t_c based on the error:

$$\text{err}(i) = \frac{\left| [\bar{u}]_{t_s}^{t_{s+i}} - [\bar{u}]_{t_s}^{t_M} \right|_{\cos}}{\left| [\bar{u}]_{t_s}^{t_M} \right|_{\cos}}$$

Then $t_c = t_s + i^*$, where i^* is the first index such that $\text{err}(i^* - h), \dots, \text{err}(i^*) < \tau$ for hold h and threshold τ .

Training state

barotropic_stirring-T85

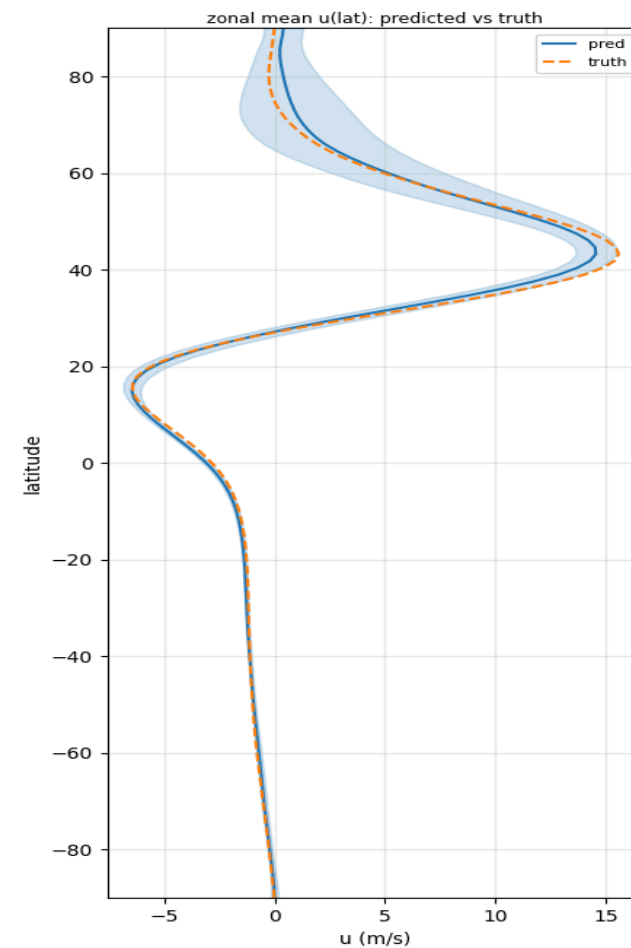
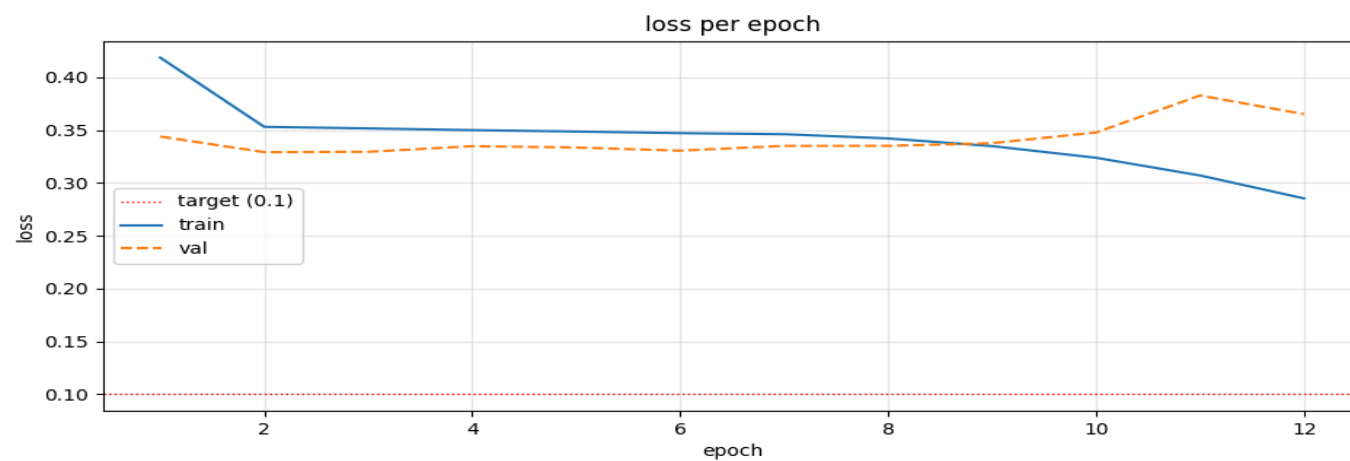
[Optimizer]
optimizer adamw
weight_decay 0.001
dropout 0.1

[Epoch]
total 100
number 12
progress 12.0%
avg duration 307.5s

[LR]
init 0.001
lr 9.65e-04
kind cosine
t_max 100
eta_min 0.0

[Loss]
kind lat_weighted_relative_l2
val 0.365197
train 0.285420
ratio 1.28
best_val 0.329210 @ ep 2

[Early Stopping]
target_loss 0.1 (val)
patience 20 ep, min_delta=0.0001 (val)



Hypothesis

Our SFNO model converges to the *mean predictor*.

```
uv run ml util baseline-score mean -c ml/configs/default.toml
{
  "loss_fn": "lat_weighted_relative_l2",
  "val_samples": 1452,
  "loss": 0.3295591851451405
}
```

Next steps: Vary stirring configuration